

CHAPTER 6

Coordination Compounds

Introduction and Werner's Theory of Coordination Compounds

- Ethylene diaminetetraacetate ion is a/an: (2024 Re)
 - hexadentate ligand
 - ambidentate ligand
 - monodentate ligand
 - bidentate ligand
- Which of the following is not an ambidentate ligand? (2024 Re)
 - $\text{C}_2\text{O}_4^{2-}$
 - SCN^-
 - NO_2^-
 - CN^-
- Given below are two statements: (2024)

Statement I: $[\text{Co}(\text{NH}_3)_6]^{3+}$ is a homoleptic complex whereas $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is a heteroleptic complex.

Statement II: Complex $[\text{Co}(\text{NH}_3)_6]^{3+}$ has only one kind of ligands but $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ has more than one kind of ligands. In the light of the above statements, choose the correct answer from the options given below:

 - Statement I is true but Statement II is false.
 - Statement I is false but Statement II is true.
 - Both Statement I and Statement II are true.
 - Both Statement I and Statement II are false.
- Homoleptic complex from the following complexes is: (2023)
 - Triamminetriaquachromium (III) chloride
 - Potassium trioxalatoaluminate (III)
 - Diamminechloridonitrito – N- platinum(II)
 - Pentaamminecarbonatocobalt (III) chloride
- Which complex compound is most stable? (2023)
 - $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$
 - $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$
 - $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$
 - $[\text{CoCl}_2(\text{en})_2]\text{NO}_3$
- Which of the following forms a set of complex and a double salt, respectively? (2023-Manipur)
 - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{CuCl}_2 \cdot 4\text{NH}_3$
 - $\text{PtCl}_2 \cdot 2\text{NH}_3$ and $\text{PtCl}_4 \cdot 2\text{HCl}$
 - $\text{K}_2\text{PtCl}_6 \cdot 2\text{NH}_3$ and $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$
 - $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{NiCl}_2(\text{H}_2\text{O})_4$
- Ethylene diaminetetraacetate (EDTA) ion is: (2021)
 - Unidentate ligand
 - Bidentate ligand with two “N” donor atoms
 - Tridentate ligand with three “N” donor atoms
 - Hexadentate ligand with four “O” and two “N” donor atoms
- The correct order of the stoichiometries of AgCl formed when AgNO_3 in excess is treated with the complexes: $\text{CoCl}_3 \cdot 6\text{NH}_3$, $\text{CoCl}_3 \cdot 5\text{NH}_3$, $\text{CoCl}_3 \cdot 4\text{NH}_3$ respectively is: (2017-Delhi)
 - 2AgCl , 3AgCl , 1AgCl
 - 1AgCl , 3AgCl , 2AgCl
 - 3AgCl , 1AgCl , 2AgCl
 - 3AgCl , 2AgCl , 1AgCl
- Cobalt(III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test for chloride ions with silver nitrate at 25°C ? (2015)
 - $\text{CoCl}_3 \cdot 4\text{NH}_3$
 - $\text{CoCl}_3 \cdot 5\text{NH}_3$
 - $\text{CoCl}_3 \cdot 6\text{NH}_3$
 - $\text{CoCl}_3 \cdot 3\text{NH}_3$
- The sum of coordination number and oxidation number of the metal M in the complex $[\text{M}(\text{en})_2(\text{C}_2\text{O}_4)]\text{Cl}$ (where en is ethylenediamine) is: (2015 Re)
 - 6
 - 7
 - 8
 - 9

Nomenclature & Isomerism in Coordination Compounds

- Match List-I with List-II. (2024)

	List-I (Complex)		List-II (Type of isomerism)
A.	$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	I.	Solvate isomerism
B.	$[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$	II.	Linkage isomerism
C.	$[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$	III.	Ionization isomerism
D.	$[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$	IV.	Coordination isomerism

Choose the correct answer from the options given below:

- A-I, B-IV, C-III, D-II
 - A-II, B-IV, C-III, D-I
 - A-II, B-III, C-IV, D-I
 - A-I, B-III, C-IV, D-II
- Type of isomerism exhibited by compounds $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$, $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ and the value of coordination number (CN) of central metal ion in all these compounds, respectively is: (2023-Manipur)
 - Geometrical isomerism, CN = 2
 - Optical isomerism, CN = 4
 - Ionisation isomerism, CN = 4
 - Solvate isomerism, CN = 6

13. Match List-I with List-II :

(2022 Re)

	List-I (Complexes)		List-II (Types)
A.	$[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$	I.	ionization isomerism
B.	$[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ and $[\text{Cr}(\text{CN})_6][\text{Co}(\text{NH}_3)_6]$	II.	coordination isomerism
C.	$[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$	III.	linkage isomerism
D.	$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$	IV.	solvate isomerism

Choose the **correct answer** from the options given below:

- a. A-IV, B-III, C-II, D-I b. A-III, B-I, C-II, D-IV
c. A-II, B-III, C-IV, D-I d. A-III, B-II, C-I, D-IV

14. The IUPAC name of the complex-

(2022)

 $[\text{Ag}(\text{H}_2\text{O})_2][\text{Ag}(\text{CN})_2]$ is:

- a. diaquasilver(I) dicyanidoargentate(I)
b. dicyanidosilver(II) diaquaargentate(II)
c. diaquasilver(II) dicyanidoargentate(II)
d. dicyanidosilver(I) diaquaargentate(I)

15. The type of isomerism shown by the complex $[\text{CoCl}_2(\text{en})_2]$ is:

(2018)

- a. Geometrical isomerism b. Coordination isomerism
c. Linkage isomerism d. Ionization isomerism

16. The name of complex ion, $[\text{Fe}(\text{CN})_6]^{3-}$ is:

(2015 Re)

- a. Hexacyanidoferrate (III) ion
b. Hexacyanoiron (III) ion
c. Hexacyanoferrate (III) ion
d. Tricyanoferrate (III) ion

17. Number of possible isomers for the complex $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ will be: (en = ethylenediamine)

(2015 Re)

- a. 4 b. 2 c. 1 d. 3

Bonding in Coordination Compounds

18. Given below are two statements:

(2024)

Statement I: Both $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.**Statement II:** $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic whereas $[\text{CoF}_6]^{3-}$ is paramagnetic.In the light of the above statements, choose the **correct** answer from the options given below:

- a. Statement I is true but Statement II is false.
b. Statement I is false but Statement II is true.
c. Both Statement I and Statement II are true.
d. Both Statement I and Statement II are false.

19. The order of energy absorbed which is responsible for the color of complexes

(2022)

- A. $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$
B. $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$ and
C. $[\text{Ni}(\text{en})_3]^{2+}$

is

- a. $B > A > C$ b. $A > B > C$
c. $C > B > A$ d. $C > A > B$

20. Match List-I with List-II.

(2021)

	List-I		List-II
(A)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(B)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(C)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(D)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the correct answer from the options given below.

- a. A-ii B-iv C-iii D-i
b. A-i B-iii C-iv D-ii
c. A-iv B-i C-ii D-iii
d. A-iv B-ii C-i D-iii

21. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds? (2020)

- a. $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$
b. $\text{F}^- < \text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
c. $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{SCN}^- < \text{F}^-$
d. $\text{SCN}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$

22. Match the coordination number and type of hybridisation with distribution of hybrid orbitals in space based on Valence bond theory.

(2020-Covid)

	Coordination number and type of hybridisation		Distribution of hybrid orbitals in space
(A)	4, sp^3	(i)	Trigonal bipyramidal
(B)	4, dsp^2	(ii)	Octahedral
(C)	5, sp^3d	(iii)	Tetrahedral
(D)	6, d^2sp^3	(iv)	Square planar

Select the correct option:

- a. A-iii B-iv C-i D-ii
b. A-iv B-i C-ii D-iii
c. A-iii B-i C-iv D-ii
d. A-ii B-iii C-iv D-i

23. What is the correct electronic configuration of the central atom in $\text{K}_4[\text{Fe}(\text{CN})_6]$ based on crystal field theory? (2019)

- a. $t_{2g}^4 e_g^2$ b. $t_{2g}^6 e_g^0$ c. $e^3 t_{2g}^3$ d. $e^4 t_{2g}^2$

24. The geometry and magnetic behaviour of the complex $[\text{Ni}(\text{CO})_4]$ are? (2018)

- a. Square planar geometry and diamagnetic
b. Tetrahedral geometry and diamagnetic
c. Tetrahedral geometry and paramagnetic
d. Square planar geometry and paramagnetic

25. Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is: (2017-Delhi)

- a. $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
b. $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
c. $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
d. $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$

26. Pick out the correct statement with respect to $[\text{Mn}(\text{CN})_6]^{3-}$ (2017-Delhi)

- It is dsp^2 hybridised and square planar
- It is sp^3d^2 hybridised and octahedral
- It is sp^3d^2 hybridised and tetrahedral
- It is d^2sp^3 hybridised and octahedral

27. The correct increasing order of trans-effect of the following species is: (2016-II)

- $\text{NH}_3 > \text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^-$
- $\text{CN}^- > \text{C}_6\text{H}_5^- > \text{Br}^- > \text{NH}_3$
- $\text{Br}^- > \text{CN}^- > \text{NH}_3 > \text{C}_6\text{H}_5^-$
- $\text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^- > \text{NH}_3$

28. Jahn-Teller effect is not observed in high spin complexes of: (2016-II)

- d^7
- d^8
- d^4
- d^9

29. The hybridisation involved in complex $[\text{Ni}(\text{CN})_4]^{2-}$ is (Atomic Number Ni = 28) (2015 Re)

- d^2sp^3
- dsp^2
- sp^3
- d^2sp^2

30. Which of these statements about $[\text{Co}(\text{CN})_6]^{3-}$ is true? (2015)

- $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a high-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration

31. Among the following complexes the one which shows zero crystal field stabilization energy (CFSE) is: (2014)

- $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$

32. A magnetic moment of 1.73 BM will be shown by one among the following: (2013)

- $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- $[\text{Ni}(\text{CN})_4]^{2-}$
- TiCl_4
- $[\text{CoCl}_6]^{4-}$

Bonding in Metal Carbonyls

33. $[\text{Mn}_2(\text{CO})_{10}]$ and $[\text{Co}_2(\text{CO})_8]$ structures have (2024 Re)

- Metal-Metal linkage
- Terminal CO groups
- Bridging CO groups
- Metal in zero oxidation state

Choose the correct answer from the options given below

- Only A, B, C
- Only B, C, D
- Only A, C, D
- Only A, B, D

34. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**. (2022 Re)

Assertion (A): The metal carbon bond in metal carbonyls possesses both σ and π character.

Reason (R): The ligand to metal bond is a π bond and metal to ligand bond is a σ bond.

In the Light of the above statements, choose the **most appropriate** answer from the options given below:

- (A) is not correct but (R) is correct
- Both (A) and (R) are correct and (R) is the correct explanation of (A)
- Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (A) is correct but (R) is not correct

35. Iron carbonyl, $\text{Fe}(\text{CO})_5$ is (2018)

- Tetranuclear
- Mononuclear
- Dinuclear
- Trinuclear

36. Which of the following has longest C-O bond length? (Free C—O bond length in CO is 1.128 Å) (2016-I)

- $[\text{Mn}(\text{CO})_6]^+$
- $\text{Ni}(\text{CO})_4$
- $[\text{Co}(\text{CO})_4]^-$
- $[\text{Fe}(\text{CO})_4]^{2-}$

Applications of Coordination Compounds

37. Which of the following complexes is used to be as an anticancer agent? (2014)

- $\text{cis-}[\text{PtCl}_2(\text{NH}_3)_2]$
- $\text{cis-}[\text{PtCl}_2\text{Br}_2]$
- $\text{Na}_2[\text{CoCl}_4]$
- $\text{mer-}[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (a) | 3. (c) | 4. (b) | 5. (d) | 6. (c) | 7. (d) | 8. (d) | 9. (d) | 10. (d) |
| 11. (c) | 12. (d) | 13. (d) | 14. (a) | 15. (a) | 16. (a) | 17. (d) | 18. (c) | 19. (d) | 20. (c) |
| 21. (d) | 22. (a) | 23. (b) | 24. (b) | 25. (b) | 26. (d) | 27. (b) | 28. (b) | 29. (b) | 30. (d) |
| 31. (a) | 32. (a) | 33. (d) | 34. (d) | 35. (b) | 36. (d) | 37. (a) | | | |